

JanSahayak AI

AI-Powered Government Document Intelligence Platform — System Architecture

1. High-Level System Architecture

USER / ADMIN Citizen questions • Profile • Document upload	REACT FRONTEND AI Q&A; • Eligibility • Summarization • Sources	FASTAPI BACKEND REST APIs • Authentication • Orchestration
DOCUMENT PROCESSING PDF/DOCX parsing • Tesseract OCR • Cleaning • Chunking	AI / RAG ENGINE Embeddings • FAISS • BM25 • Hybrid Retrieval • LLM	ELIGIBILITY ENGINE Profile matching • Criteria evaluation • Evidence
NLP / ML Document Classification • spaCy NER • Metadata Extraction	DATA LAYER PostgreSQL metadata • FAISS vector index • Document store	OUTPUT Grounded answer • Summary • Eligibility result • Citation

Deployment: Docker + Docker Compose. The LLM can run locally through Ollama with an open-source model; Gemini API can be used optionally for comparison/testing.

2. Document Ingestion & Indexing Pipeline

Step	Component	What happens
1	Official Government Documents	Collect official PDFs/DOCX/XLSX from government portals or upload through the Admin interface.
2	Text Extraction	Extract selectable text from normal PDFs and documents.
3	OCR	Use Tesseract OCR when a PDF is scanned/image-based.
4	Cleaning & Chunking	Clean text and divide long documents into meaningful chunks while retaining page/document metadata.
5	ML/NLP	Classify document type and extract entities such as schemes, dates, amounts, organizations and locations.
6	Embeddings	Convert text chunks into semantic vectors using Sentence Transformers.
7	Indexing	Store vectors in FAISS and document metadata in PostgreSQL.

3. User Question → RAG → Answer Pipeline

User Question Natural-language query	Query Processing Create query embedding	Hybrid Retrieval FAISS semantic search + BM25 keyword search	Top Relevant Chunks Rank and select evidence	RAG + LLM Generate grounded response	Answer Response + document + page citation
--	---	---	--	--	--

4. Eligibility Matching Pipeline

User Profile	Scheme Criteria	Matching Engine	Result
Age • Education • State • Income • Other profile fields	Retrieved official eligibility rules	Rules/criteria matching with evidence from the source documents	Potentially eligible / Not eligible + reasons + citation

5. Automatic Policy Summarization Pipeline

Long Government Document	Text Extraction + Chunking	LLM / Summarization	Structured Summary Objective • Eligibility • Benefits • Requirements • Key information
--------------------------	----------------------------	---------------------	--

6. Technology Stack

Layer	Technology
Frontend	React.js + Tailwind CSS
Backend	Python + FastAPI + Uvicorn
Document Processing	PyMuPDF / python-docx + Tesseract OCR
NLP / ML	Sentence Transformers + scikit-learn + spaCy
Search / RAG	FAISS + BM25 + LangChain or LlamaIndex
LLM	Ollama + open-source model (e.g., Qwen/Llama/Gemma); optional Gemini API
Database	PostgreSQL
Deployment	Docker + Docker Compose